Complete cold DLP runtime through ecbench

Run 20261019-174315-boolean-final-ecbench-25cc20, source 71c4c99d85. Complete one-target cold DLP recovery on dedicated c7i.2xlarge; native ecbench L2 requirement, one worker,8 independent public targets per curve,5 measured rounds plus1 warmup,11 interleaved arms. Sources/specs and complete audit certificates remain adjacent/raw. Missing, failed and below-level rows are retained; a launch is not a result.

All methods within each session share the same public targets and standard polynomial subspace: dimension5 at n17 and6 at n23. These bases differ from the component experiment's Frobenius-invariant bases. The curve identities are icv1-f2m17-tm101-00378d4e and icv1-f2m23-tm5197-1f85e9e1. This two-size panel is not a scaling claim.

The pipeline solve clock includes factor-base construction, oracle prepare, relation collection, linear algebra, target descent/recovery and internal verification. Whole child process additionally includes adapter construction, report metadata, common workload construction and startup. Both clocks are recorded; the native harness's comparison gate uses the pipeline solve clock. Independent parent verification/ replay, compilation and warmup are outside these clocks.

F4/F5 baseline uses the prior recursive policy with qualified GF2 kernels. Sources0acfc55b7f/71c4c99d85 use candidate matrix_runtime=quadratic-tail; corrected source9bed104b0a uses adaptive-quadratic-tail, with a different method identity. F6 names kernel_runtime=native-v1. Generalized typed roots and both Semaev table backends participate. Environment-override refusal remains intact; no admission threshold is relaxed.

Wall ratios below are B/A from ecbench's own paired bootstrap and A/A gate. Admission, completeness and outside_noise must all support a gain; partial/failed sessions do not become successful comparisons. Process medians are supplementary observed clocks, not a replacement gate.

Counted S, generic floor and matched strong rho are retained in table-n*.txt and ecbench-summary.json. Gray assignments are explicitly counted unpriced; reference native unpriced counters are retained too. Sensitivity bands at0/0.1/1 GAE per recorded operation are diagnostic and deduplicate generalized solver-count aliases. An unquantified counter stays unquantified. No ratio of lower bounds is a full-cost ranking.

Exact original-output and replay checks, query-count comparisons, source/binary hashes, failures and job resources remain in the evidence. All-L2 coverage is separate from execution/verification completeness; a fully verified session can have below-level timing rows. Low-sample clocks are descriptive; no threshold is relaxed. New matrix policies stay opt-in; no external-fastest, other-hardware or rho-beating claim is made.

n17: 528 total records, 88 warmups; replay audit OK True; all requested L2 timing coverage False.

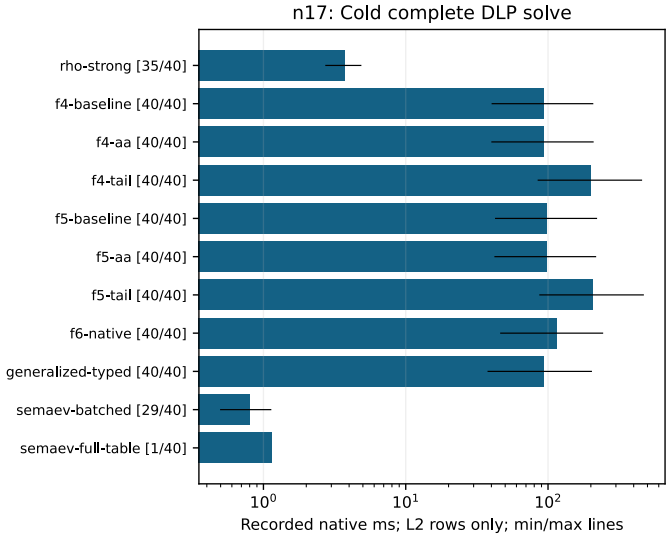
n17 f4: candidate/baseline pipeline wall 2.1615, CI95 [2.1395826336351726, 2.1927699402827145], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.

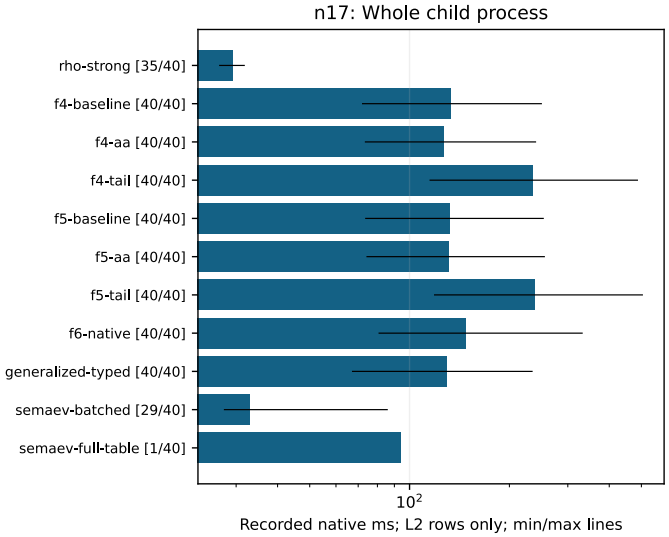
n17 f5: candidate/baseline pipeline wall 2.1155, CI95 [2.0821687846214223, 2.133465917871537], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.

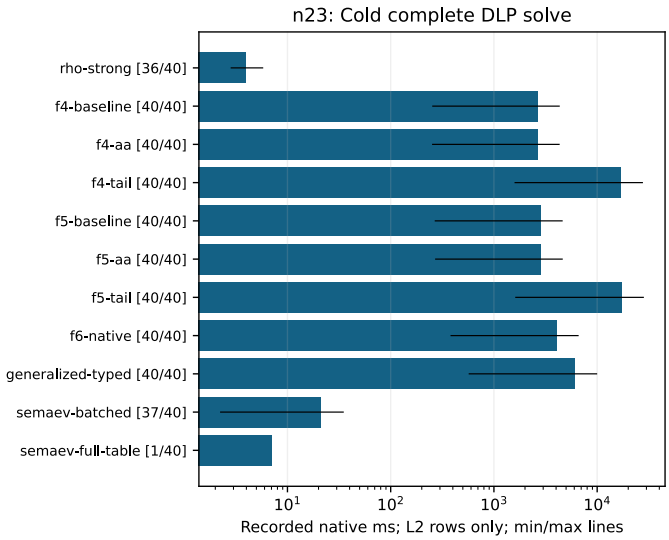
n23: 528 total records, 88 warmups; replay audit OK True; all requested L2 timing coverage False.

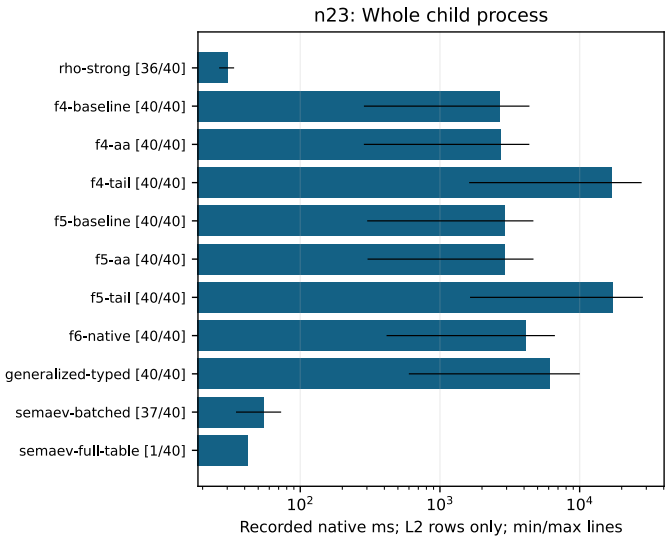
n23 f4: candidate/baseline pipeline wall 6.3963, CI95 [6.374242969640284, 6.412638088211264], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.

n23 f5: candidate/baseline pipeline wall 6.0804, CI95 [6.063029417443889, 6.086417851352189], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.


n17: Cold complete DLP solve
rho-strong [35/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [29/40]
semaev-full-table [1/40]
10^0
10^1
10^2
Recorded native ms; L2 rows only; min/max lines


n17: Whole child process
rho-strong [35/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [29/40]
semaev-full-table [1/40]
10^2
Recorded native ms; L2 rows only; min/max lines


n23: Cold complete DLP solve
rho-strong [36/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [37/40]
semaev-full-table [1/40]
10^1
10^2
10^3
10^4
Recorded native ms; L2 rows only; min/max lines


n23: Whole child process
rho-strong [36/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [37/40]
semaev-full-table [1/40]
10^2
10^3
10^4
Recorded native ms; L2 rows only; min/max lines

Recorded clocks and admission

n	Arm	Solve ms	Process ms	Verified	L2	Requested	All L2
17	rho-strong	3.7365	29.3152	40	35	40	False
17	f4-baseline	93.1034	132.9379	40	40	40	True
17	f4-aa	93.1480	126.7359	40	40	40	True
17	f4-tail	199.4259	235.4926	40	40	40	True
17	f5-baseline	97.8361	132.1757	40	40	40	True
17	f5-aa	98.3205	131.2446	40	40	40	True
17	f5-tail	205.3710	238.4111	40	40	40	True
17	f6-native	114.7783	147.3245	40	40	40	True
17	generalized-typed	93.0190	128.9489	40	40	40	True
17	semaev-batched	0.8028	32.8160	40	29	40	False
17	semaev-full-table	1.1417	94.1875	40	1	40	False
23	rho-strong	3.9677	30.2493	40	36	40	False
23	f4-baseline	2678.9874	2711.1398	40	40	40	True
23	f4-aa	2681.4155	2713.8800	40	40	40	True
23	f4-tail	17138.2669	17170.6120	40	40	40	True
23	f5-baseline	2873.5406	2907.5252	40	40	40	True
23	f5-aa	2862.7236	2896.9232	40	40	40	True
23	f5-tail	17373.4405	17407.0452	40	40	40	True
23	f6-native	4067.4085	4099.9347	40	40	40	True
23	generalized-typed	6126.6127	6159.9945	40	40	40	True
23	semaev-batched	21.2930	54.8103	40	37	40	False
23	semaev-full-table	7.0579	41.9045	40	1	40	False

Native harness comparisons; B/A

n	Pair	Wall gate	Median ratio	CI95	Outside noise	Below L2
17	f4	admitted	2.1615	2.1396..2.1928	True	0
17	f5	admitted	2.1155	2.0822..2.1335	True	0
17	rho-f4-tail	admitted	51.9373	46.9349..65.1904	True	5
17	rho-f5-tail	admitted	53.6985	48.2097..67.3411	True	5
17	rho-f6-native	admitted	29.4145	24.9855..37.1320	True	5
17	rho-generalized-typed	admitted	23.8976	20.2568..30.4485	True	5
17	rho-semaev-batched	admitted	0.2082	0.1850..0.2437	True	14
17	rho-semaev-full-table	descriptive	0.2870	missing	None	39
23	f4	admitted	6.3963	6.3742..6.4126	True	0
23	f5	admitted	6.0804	6.0630..6.0864	True	0
23	rho-f4-tail	admitted	4714.9990	3320.1775..5394.5115	True	4
23	rho-f5-tail	admitted	4794.1400	3365.9942..5423.8279	True	4
23	rho-f6-native	admitted	1124.1737	793.9122..1270.3334	True	4
23	rho-generalized-typed	admitted	1688.8656	1206.0619..1922.7368	True	4
23	rho-semaev-batched	admitted	6.1800	4.3536..6.6396	True	6
23	rho-semaev-full-table	descriptive	4.9896	missing	None	40

Partly priced S sensitivity; all entries diagnostic

n	Arm	0 GAE/op	0.1 GAE/op	1 GAE/op	Lower bound	Unknown entries
17	rho-strong	4.0416	4.185	5.4753	True	0
17	f4-baseline	43.799	43.799	43.799	False	0
17	f4-aa	43.799	43.799	43.799	False	0
17	f4-tail	143.67	143.74	144.29	True	0
17	f5-baseline	43.731	43.731	43.731	False	0
17	f5-aa	43.731	43.731	43.731	False	0
17	f5-tail	142.99	143.05	143.6	True	0
17	f6-native	33.082	33.082	33.082	False	0
17	generalized-typed	16.421	1080.1	10653	True	0
17	semaev-batched	17.317	17.317	17.317	False	0
17	semaev-full-table	18.083	18.083	18.083	False	0
23	rho-strong	0.88552	0.95031	1.5335	True	0
23	f4-baseline	72.251	72.251	72.251	False	0
23	f4-aa	72.251	72.251	72.251	False	0
23	f4-tail	1138.9	1138.9	1139.2	True	0
23	f5-baseline	72.228	72.228	72.228	False	0
23	f5-aa	72.228	72.228	72.228	False	0
23	f5-tail	1137.1	1137.2	1137.5	True	0
23	f6-native	65.541	65.541	65.541	False	0
23	generalized-typed	43.007	16897	1.6859e+05	True	0
23	semaev-batched	43.709	43.709	43.709	False	0
23	semaev-full-table	44.039	44.039	44.039	False	0

Explicit unpriced operation ledgers; different units are not equated

n	Arm	Unit	Samples	Median count	Min	Max
17	rho-strong	canonicalisations_uncharged	40	192.0	64	356
17	rho-strong	partition_hashes_uncharged	40	148.5	30	306
17	rho-strong	table_inserts_uncharged	40	9.0	2	20
17	rho-strong	table_queries_uncharged	40	10.0	3	22
17	f4-tail	quadratic_tail_assignments_uncharged	40	132.0	4	426
17	f5-tail	quadratic_tail_assignments_uncharged	40	132.0	4	426
17	generalized-typed	generalized_assignments_uncharged	40	2613760.0	1051648	5709824
23	rho-strong	canonicalisations_uncharged	40	672.0	160	1217
23	rho-strong	partition_hashes_uncharged	40	594.5	119	1105
23	rho-strong	table_inserts_uncharged	40	39.5	12	79
23	rho-strong	table_queries_uncharged	40	40.5	13	80
23	f4-tail	quadratic_tail_assignments_uncharged	40	599.0	12	2000
23	f5-tail	quadratic_tail_assignments_uncharged	40	599.0	12	2000
23	generalized-typed	generalized_assignments_uncharged	40	355270656.0	33546240	580587520

Admitted phase clocks; recorded ms

n	Arm	Phase	Median ms	Min ms	Max ms	Samples
17	f4-baseline	factor_base	0.1426	0.1345	0.1710	40
17	f4-baseline	linear_algebra	0.0121	0.0041	0.0225	40
17	f4-baseline	oracle_setup	0.0509	0.0486	0.0668	40
17	f4-baseline	relations	92.8382	39.8707	207.8025	40
17	f4-baseline	verify	0.0020	0.0017	0.0095	40
17	f4-aa	factor_base	0.1429	0.1373	0.2398	40
17	f4-aa	linear_algebra	0.0125	0.0038	0.0232	40
17	f4-aa	oracle_setup	0.0508	0.0481	0.0612	40
17	f4-aa	relations	92.8854	39.7144	208.9281	40
17	f4-aa	verify	0.0021	0.0016	0.0105	40
17	f4-tail	factor_base	0.1438	0.1372	0.1653	40
17	f4-tail	linear_algebra	0.0117	0.0038	0.0199	40
17	f4-tail	oracle_setup	0.0513	0.0477	0.0597	40
17	f4-tail	relations	199.1587	84.2992	458.0556	40
17	f4-tail	verify	0.0021	0.0017	0.0069	40
17	f5-baseline	factor_base	0.1426	0.1353	0.2570	40
17	f5-baseline	linear_algebra	0.0108	0.0060	0.0221	40
17	f5-baseline	oracle_setup	0.0515	0.0483	0.0649	40
17	f5-baseline	relations	97.5765	42.1526	220.9630	40
17	f5-baseline	verify	0.0021	0.0017	0.0099	40
17	f5-aa	factor_base	0.1422	0.1329	0.1568	40
17	f5-aa	linear_algebra	0.0158	0.0064	0.0207	40
17	f5-aa	oracle_setup	0.0509	0.0479	0.0935	40
17	f5-aa	relations	98.0624	41.7243	217.6222	40
17	f5-aa	verify	0.0021	0.0016	0.0096	40
17	f5-tail	factor_base	0.1431	0.1351	0.1612	40
17	f5-tail	linear_algebra	0.0124	0.0036	0.0244	40
17	f5-tail	oracle_setup	0.0510	0.0490	0.0772	40
17	f5-tail	relations	205.0939	86.5878	471.3343	40
17	f5-tail	verify	0.0021	0.0017	0.0112	40
17	f6-native	factor_base	0.1432	0.1346	0.2701	40
17	f6-native	linear_algebra	0.0134	0.0034	0.0247	40
17	f6-native	oracle_setup	0.0493	0.0459	0.0786	40
17	f6-native	relations	114.5068	45.7136	243.7838	40
17	f6-native	verify	0.0021	0.0016	0.0122	40
17	generalized-typed	factor_base	0.1423	0.1357	0.1720	40
17	generalized-typed	linear_algebra	0.0149	0.0044	0.0206	40
17	generalized-typed	oracle_setup	0.0477	0.0451	0.1339	40
17	generalized-typed	relations	92.7673	37.3378	203.2972	40
17	generalized-typed	verify	0.0021	0.0016	0.0086	40
17	semaev-batched	factor_base	0.1484	0.1424	0.1738	29
17	semaev-batched	linear_algebra	0.0089	0.0041	0.0132	29
17	semaev-batched	oracle_setup	0.0157	0.0143	0.0184	29
17	semaev-batched	relations	0.5867	0.2946	0.9212	29
17	semaev-batched	verify	0.0021	0.0016	0.0032	29
17	semaev-full-table	factor_base	0.1716	0.1716	0.1716	1
17	semaev-full-table	linear_algebra	0.0120	0.0120	0.0120	1
17	semaev-full-table	oracle_setup	0.3119	0.3119	0.3119	1
17	semaev-full-table	relations	0.6013	0.6013	0.6013	1
17	semaev-full-table	verify	0.0021	0.0021	0.0021	1
23	f4-baseline	factor_base	0.1597	0.0643	0.1939	40
23	f4-baseline	linear_algebra	0.0333	0.0055	0.0566	40
23	f4-baseline	oracle_setup	0.0796	0.0755	0.0951	40
23	f4-baseline	relations	2678.7347	252.3234	4340.5816	40
23	f4-baseline	verify	0.0039	0.0033	0.0179	40
23	f4-aa	factor_base	0.1626	0.0610	0.2657	40
23	f4-aa	linear_algebra	0.0298	0.0061	0.0583	40
23	f4-aa	oracle_setup	0.0796	0.0742	0.0951	40
23	f4-aa	relations	2681.1132	251.3995	4331.1143	40
23	f4-aa	verify	0.0039	0.0033	0.0157	40
23	f4-tail	factor_base	0.1628	0.0679	0.1948	40
23	f4-tail	linear_algebra	0.0421	0.0049	0.1207	40
23	f4-tail	oracle_setup	0.0807	0.0757	0.0899	40
23	f4-tail	relations	17137.9602	1583.1918	27752.8807	40
23	f4-tail	verify	0.0040	0.0034	0.0182	40
23	f5-baseline	factor_base	0.1614	0.0634	0.1992	40

n	Arm	Phase	Median ms	Min ms	Max ms	Samples
23	f5-baseline	linear_algebra	0.0323	0.0058	0.0510	40
23	f5-baseline	oracle_setup	0.0798	0.0749	0.0923	40
23	f5-baseline	relations	2873.2932	266.6973	4632.5200	40
23	f5-baseline	verify	0.0039	0.0034	0.0187	40
23	f5-aa	factor_base	0.1614	0.0633	0.1964	40
23	f5-aa	linear_algebra	0.0315	0.0098	0.0526	40
23	f5-aa	oracle_setup	0.0804	0.0748	0.1979	40
23	f5-aa	relations	2862.4205	269.0584	4641.4401	40
23	f5-aa	verify	0.0041	0.0032	0.0176	40
23	f5-tail	factor_base	0.1612	0.0647	0.1864	40
23	f5-tail	linear_algebra	0.0355	0.0089	0.0615	40
23	f5-tail	oracle_setup	0.0789	0.0753	0.0912	40
23	f5-tail	relations	17373.1240	1608.3349	28340.2713	40
23	f5-tail	verify	0.0040	0.0034	0.0184	40
23	f6-native	factor_base	0.1601	0.0641	0.2976	40
23	f6-native	linear_algebra	0.0369	0.0055	0.0637	40
23	f6-native	oracle_setup	0.0768	0.0728	0.1028	40
23	f6-native	relations	4067.1103	379.4825	6615.5331	40
23	f6-native	verify	0.0040	0.0036	0.0198	40
23	generalized-typed	factor_base	0.1607	0.0653	0.3056	40
23	generalized-typed	linear_algebra	0.0310	0.0049	0.0610	40
23	generalized-typed	oracle_setup	0.0751	0.0718	0.0872	40
23	generalized-typed	relations	6126.3234	569.2020	9997.0456	40
23	generalized-typed	verify	0.0039	0.0032	0.0051	40
23	semaev-batched	factor_base	0.1698	0.1621	0.2594	37
23	semaev-batched	linear_algebra	0.0194	0.0033	0.0347	37
23	semaev-batched	oracle_setup	0.0292	0.0275	0.0368	37
23	semaev-batched	relations	21.0244	1.9585	34.7290	37
23	semaev-batched	verify	0.0039	0.0032	0.0105	37
23	semaev-full-table	factor_base	0.0745	0.0745	0.0745	1
23	semaev-full-table	linear_algebra	0.0091	0.0091	0.0091	1
23	semaev-full-table	oracle_setup	0.4131	0.4131	0.4131	1
23	semaev-full-table	relations	6.5128	6.5128	6.5128	1
23	semaev-full-table	verify	0.0034	0.0034	0.0034	1